Lecture 1 and lecture 2

Answers to the Practice Exercises can be found in the back of this ebook.

1. Which label accurately describes neurons that detect information from the physical world and communicate that information along to the central nervous system?

1. motor neuron

2 sensory neuron

3. interneuron

4. glia

2. Parkinson's disease is associated with the loss of neurons that produce which of the following neurotransmitters?

1. acetylcholine

2. norepinephrine

3. dopamine

4. serotonin

3. Drugs can produce the following actions on neurotransmitter activity. Label each example as either an agonist or antagonist effect.

1. mimic the neurotransmitter and activate the postsynaptic receptor

2. block the reuptake of the neurotransmitter

3. decrease neurotransmitter release from the presynaptic neuron

4. clear the neurotransmitter from the synapse

4. Which of the following statements about behavioral genetics is false?

1. Heritability refers to traits passed from parent to offspring.

2. Similarities among nonbiological adopted siblings are inferred to reflect environmental influences.

3. Identical twins raised apart are often more similar than identical twins raised together.

4. Greater similarities between monozygotic twins compared with dizygotic twins are inferred to reflect genetic influences.

5. What is the role of a neuron's membrane in an action potential?

1. It is a fatty layer that insulates the axon.

2. It secretes neurotransmitters to communicate with surrounding neurons.

3. It absorbs nutrients.

4. It contains ion channels that regulate electrical potential and firing.

6. Which of the following techniques can provide information about whether a particular brain region is necessary for a task?

1. electroencephalography (EEG)

2. functional magnetic resonance imaging (fMRI)

3. positron emission tomography (PET)

4. transcranial magnetic stimulation (TMS)

7. Which statement about split-brain patients is true?

1. They have had surgery to therapeutically remove one hemisphere of the brain.

2. The left hemisphere can perceive stimuli, but the right hemisphere cannot.

3. The left hemisphere can verbally report its perception. The right hemisphere cannot articulate what it saw but can act on its perception.

4. The left hemisphere is analytical, and the right hemisphere is creative.

8. When a neuron is at rest, how does the electrical charge inside the cell compare with the charge outside?

1. The outside and the inside are equivalent in charge.

2. The outside is more negatively charged than the inside.

3. The inside is more negatively charged than the outside.

4. It varies depending on whether it is a presynaptic or postsynaptic neuron.

9. Someone suffers a stroke that causes damage to the left motor cortex. Which of the following impairments will the person most likely exhibit?

1. an inability to comprehend spoken language

2. an inability to recognize faces

3. paralysis of the left side of the body

4. paralysis of the right side of the body

10. What pair of words best matches genotype and phenotype, respectively?

1. underlying and observed

2. observed and underlying

3. dominant and recessive

4. recessive and dominant

Lecture 3 consiousness:

1. What is a key distinction between a person in an unresponsive wakefulness state and a person in a minimally conscious state?

1. The person in the minimally conscious state is less responsive to their surroundings.

2. The person in the unresponsive wakefulness state is less likely to regain full consciousness at some point in the future.

3. The person in the minimally conscious state is dreaming, whereas the person in the unresponsive wakefulness state is in a coma.

2. A researcher subliminally flashes the word thirst to participants in Condition A, and nothing to participants in Condition B. Both groups of participants are then given the opportunity to drink some water. What do you predict about the participants' behavior?

1. Participants in Conditions A and B will behave nearly identically.

2. Participants in Condition A will drink more water than participants in Condition B, especially if they are thirsty.

3. Participants in Condition B will drink more water than participants in Condition A, especially if they are thirsty.

3. Match each definition with one of the following attention terms: change blindness, automatic processing, controlled processing, shadowing.

1. well-practiced tasks that can be done quickly, with little conscious effort

2. repeating the auditory input to one ear while different input is transmitted to the other ear

3. failure to notice large changes in one's environment

4. difficult or challenging tasks that require conscious effort

4. For each description below, name the sleep disorder: insomnia, apnea, narcolepsy, somnambulism.

1. Marcus falls asleep suddenly while practicing piano.

2. Emma walks through the living room in the middle of the night, seemingly oblivious to those around her.

3. Sophia spends most of the night trying to fall asleep and is anxious and has a hard time functioning during the day.

4. Ivan snores loudly and stops breathing for short periods while sleeping.

5. Which of the following pieces of evidence does not support the idea that sleep is an adaptive behavior?

1. Sleep allows the body and brain to restore themselves.

2. Some animals, such as some frogs, do not sleep.

3. Humans sleep at night because our ancestors were more at risk in the dark.

4. Students who study more tend to have more REM sleep.

6. Which of the following instruction sets would a yoga teacher trained in concentrative meditation be most likely to give?

1. “Close your eyes while sitting in a comfortable position. Let your thoughts move freely through your mind, like clouds passing through the sky. Acknowledge them, but do not react to them.”

2. “Lying on your back, rest your hands gently on your abdomen. As you breathe in and out, focus attention on your breath. Notice the rhythmic rise and fall of your abdomen and the slow, deep movement of your chest.”

3. “Standing in place, bend one knee and lift that leg. Grasp the foot and bring it back as far as possible. Then lower the foot and repeat this action with the other knee, leg, and foot.”

7. Which of the following drugs is classified as a stimulant?

1. marijuana

2. cocaine

3. heroin

4. alcohol

5. LSD

8. Which of the following is an example of exogenous attention?

1. You focus your attention on reading the textbook.

2. You attend to the road while driving.

3. You attend to an unexpected, sudden loud noise.

9. Match each of the following drugs with the appropriate description of a user’s typical response: alcohol, marijuana, MDMA, opiates, stimulants.

1. increased heart rate, elevated mood, restlessness

2. relaxation, contentment, vivid perceptual experiences

3. impaired motor skills, decreased sexual performance

4. energy, slight hallucinations

5. lack of pain, euphoria, intense pleasure

10. Which statement is false in regard to REM sleep?

1. You are more likely to report dreaming if woken from REM sleep.

2. You are more likely to sleepwalk during REM sleep.

3. As morning approaches, the sleep cycle becomes shorter and the sleeper spends relatively more time in REM sleep.

4. Most of the body’s muscles are paralyzed during REM sleep

Lecture 4:

1. Which answer accurately lists the order in which these structures participate in sensation and perception (except for smell)?

1. thalamus, specialized receptors, cortex

2. specialized receptors, cortex, thalamus

3. cortex, specialized receptors, thalamus

4. specialized receptors, thalamus, cortex

2. Match each process of detecting and interpreting information— sensation, perception, bottom-up processing, top-down processingwith its definition.

1. processing based on prior knowledge and expectation

2. detection of sensory signals by receptors

3. processing and interpretation of sensory signals

4. perception based on physical features of sensory stimuli

3. Which sense organ is largest in humans?

1. eyes, due to the large number of cones densely packed in the fovea

2. ears, due to the curvature of the cochlea, which increases surface area of the basilar membrane to house an infinite number of hair cells

3. nose, due to the dense array of cells packed within the olfactory epithelium

4. tongue, due to the large number of taste buds that can be housed within each papilla

4. skin, due to the large surface area 4. In audition, detecting the _____ of the sound wavelength results in the perception of loudness. Detecting the _____ of the wavelength results in the perception of pitch.

1. frequency, amplitude

2. amplitude, frequency

3. frequency, hertz

4. hertz, frequency

5. Identify each of the following visual perceptions as an example of size constancy, shape constancy, color constancy, or lightness constancy.

1. recognizing a dinner plate as circular, even when viewing it at an angle makes it appear elliptical

2. labeling grass as green, even in the dark

3. identifying a building as a skyscraper, even though it appears smaller than other objects in your field of vision

4. recognizing a door as a door, even when it is fully open so that you see only the edge

5. knowing that your friend is wearing the same T-shirt when it is illuminated differently by bright fluorescent lights or dull lamp lights

6. Imagine you have a dull, chronic pain across your lower back. No matter how you position yourself, you cannot make the pain go away. Which combination of descriptors is most related to the signaling of this type of pain?

1. activated by chemical changes in tissue, fast fibers, nonmyelinated axons

2. activated by strong physical pressure or temperature extremes, fast fibers, myelinated axons

3. activated by chemical changes in tissue, slow fibers, nonmyelinated axons

4. activated by strong physical pressure or temperature extremes, slow fibers, myelinated axons

7. After hours of waiting, a bird-watcher hears the call of a rare bird species. The bird-watcher reacts by turning their head about 45 degrees to the left and sees the bird. Which of the following statements best describes how the bird-watcher knew where to turn their head?

1. The call reached the bird-watcher's left ear before the right ear; the call was less intense in the bird-watcher's right ear than in the left ear.

2. The call reached the bird-watcher's left ear before the right ear; the call was less intense in the bird-watcher's left ear than in the right ear.

3. The call reached the bird-watcher's right ear before the left ear; the call was less intense in the bird-watcher's right ear than in the left ear.

4. The call reached the bird-watcher's right ear before the left ear; the call was less intense in the bird-watcher's left ear than in the right ear.

8. In which lobe of the brain (frontal, occipital, parietal, temporal) does each of the following sensory cortices reside?

1. primary auditory cortex

2. primary somatosensory cortex

3. primary visual cortex

9. Which of the following statements about the gustatory system are true?

1. Taste is a mixture of five basic kinds of gustatory receptors.

2. Different regions of the tongue are more sensitive to different tastes.

3. The texture of a food can affect how its taste is perceived.

4. People lose half of their taste buds by age 20.

5. Supertasters tend to be overweight and are more likely to be male.

10. Imagine you are preparing to conduct a brain imaging study of visual processing. Which pathway—dorsal or ventral—do you hypothesize will be activated by each of the following experimental tasks?

1. deciding which of two objects is farther away

2. describing an object's color

3. describing a silhouette's shape

4. naming an object

5. selecting which two of three objects are closest together

Lecture 5

1. On the first day of summer school, the air conditioner in Matt's history classroom broke down. All the other rooms were unaffected. After several weeks of attending class in the sweltering room, Matt started sweating every time he approached the history room. In this example of classical conditioning, what are the US, UR, CS, and CR?

2. Which of the following statements is true about extinction learning?

1. It involves fewer presentations of the CS than during acquisition.

2. It permanently eliminates the expression of the conditioned response.

3. It is a form of new learning that replaces the associative bond between the CS and US but does not eliminate it.

3. During _____, organisms become less responsive to stimuli; during _____, organisms become more responsive to stimuli.

1. tolerance, sensitization

2. habituation, sensitization

3. sensitization, habituation

4. addiction, dependence

4. Jason just moved his dog's food to the cupboard, which creaks noisily when it is opened. In which situation is Jason's dog likely to learn an association between the sound of the cupboard opening and the food?

1. The door reliably opens shortly after the food is delivered.

2. The door reliably opens shortly before the food is delivered.

3. The door reliably opens just as the food is delivered.

4. The door reliably opens along with the sound of a can opener, which has previously signaled food delivery.

5. Identify each statement as an example of negative punishment, positive punishment, negative reinforcement, or positive reinforcement.

1. Whenever a puppy barks, it gets its belly rubbed, so it barks more.

2. A teenager who stays out too late is given extra chores to do.

3. A person with a clean driving record receives a reduced insurance premium.

4. Your date arrives an hour late, and you refuse to speak to them for the rest of the evening.

6. Match each scenario with one of the following reinforcement schedules: fixed ratio, variable ratio, fixed interval, variable interval.

1. Your mail arrives at the same time every day.

2. Every once in a while, you receive exciting mail.

3. Every seventh newspaper you receive is loaded with coupons.

4. You buy a packet of baseball cards each week. Sometimes the packet includes a valuable card, but other times it does not.

7. Match each example with the type of learning it describes: nonassociative, associative, social.

1. Maya finds that she is speaking up more in class after her professor compliments her on the questions she asks.

2. Donae no longer responds to a persistent beep from a broken appliance in his apartment.

3. After watching a YouTube video, Rohan is able to change the air filter in his heater.

8. Which of the following statements are true?

1. Any behavior can be shaped through reinforcement.

2. Psychologists believe that punishment is more effective than positive reinforcement for shaping behavior.

3. Animals can be conditioned to fear flowers as easily as they can be conditioned to fear snakes.

4. A conditioned taste aversion can be learned in a single trial.

9. Why is it important for addiction treatment to include exposure to conditioned environmental drug cues?

10. Which statement correctly describes the relationship between dopamine and reinforcement?

1. Dopamine is more important for the “liking” rather than the “wanting” aspect of reward.

2. Dopamine activity in the brain is increased when a behavior leads to an unexpected reward.

3. Only primary reinforcers are linked with dopamine activity

Lecture 6:

1. What is the sequence of memory stages?

1. storage (consolidation) → encoding → retrieval

2. encoding → retrieval → storage (consolidation)

3. retrieval → storage (consolidation) → encoding

4. encoding → storage (consolidation) → retrieval

2. Which of the following phenomena increase the likelihood of forgetting? Choose all that apply.

1. shallow encoding

2. elaborative encoding

3. blocking

4. proactive interference

5. retroactive interference

6. chunking

3. All the details of flashbulb memories are always more accurate than those of normal memories. True or false?

4. Which finding(s) supports the conclusion that working memory and long-term memory are distinct memory processes? Choose all that apply.

1. Patient H.M. retained working memory without being able to form new long-term memories.

2. The primacy effect requires long-term memory, whereas the recency effect requires working memory.

3. The primacy effect requires working memory, whereas the recency effect requires long-term memory.

4. choices a and b

5. choices a and c

5. How can capacity of working memory be increased?

1. blocking

2. maintenance rehearsal

3. chunking

4. reconsolidation

6. Imagine you are a manager seeking to hire a new employee. Before leaving work Monday afternoon, you look through a stack of 30 resumes organized alphabetically by last name. When you return to work Tuesday morning, which job applicant are you most likely to remember?

1. Alvarado, because of the primacy effect

2. Martonosi, because of persistence

3. Russo, because of reconsolidation

4. Zambino, because of the recency effect

7. Which statement is accurate regarding effective study strategies when preparing for an examination?

1. Maintenance rehearsal is the most effective way to deepen encoding.

2. Six 1-hour study sessions are more effective than one 6-hour study session.

3. Elaborative rehearsal is likely to lead to blocking.

4. Practiced retrieval will increase retroactive interference.

8. You ask a friend to memorize the following list: bed, rest, night, tired, blanket, pillow, relaxed. Later, you ask your friend to tell you as many words from the list as they can remember. In addition to remembering some of the words, your friend lists the word sleep, which did not appear on the original list. Which of the following phenomena is most related to your friend's error in recall?

1. context-dependent encoding

2. culturally relevant schemas

3. false memories

4. state-dependent encoding

9. Which memory flaw is demonstrated in each of the following examples?

1. A friend introduces you to her brother; 5 minutes later, you find you cannot remember the brother's name.

2. You see someone on a bus whom you think you recognize from high school. Later you realize he is familiar because he appeared in a TV commercial you have seen.

10. Fill in the blanks. People experiencing _____ amnesia are unable to recall memories from the past, whereas people experiencing _____ amnesia are unable to form new memories.

Lecture 7 and 8:

1. Identify each of the following examples as either an analogical representation, a symbolic representation, or both.

1. the written word cat

2. a cartoon illustration of a cat

3. a mental map of the United States

4. the spoken word America

2. Match each of the following decisions with the heuristic used: anchoring, framing, availability heuristic, representativeness heuristic.

1. Magda chooses not to travel, assuming she is at high risk for contracting the Zika virus, after seeing frequent news reports about it.

2. Jacob decides to purchase flood insurance, motivated by concern for the potential loss of his valuable property.

3. Steven's doctor is not concerned by Steven's chest pain because he is thin and young, so does not fit the typical risk profile for heart attack.

4. Sylvia decides to purchase 10 oranges, rather than her usual 6, because they are being sold "10 for $10."

5. Which of the following statements about brain regions and language processing is correct?

1. Broca's area controls speech comprehension.

2. Wernicke's area controls speech production.

3. The amygdala controls the acquisition of vocabulary.

4. The left hemisphere primarily controls speech production and comprehension.

6. Casey is very empathic. He is able to accurately interpret what people are thinking and feeling, if someone is uncomfortable, stressed, or attracted to something. Based on these abilities, which measure of intelligence would Casey score well on?

1. crystallized intelligence

2. EI

3. fluid intelligence

4. g

7. Which of the following characteristics is not typically correlated with high scores on intelligence tests?

1. faster reaction times, particularly in a choice reaction time test

2. increased likelihood of premature death

3. increased working memory performance

4. a stimulating, enriched environment

Lecture 9

1.Which of the following jobs most likely involves producer-ofresearch skills rather than consumer-of-research skills?

1. Police officer
2. University professor
3. Physician
4. Journalist

2. As a true empiricist, one should:

1. Base one's conclusions on direct observations.

2. Strive to prove all theories with research.

3. Be sure that one's research can be applied in a real-world setting.

4. Discuss one's ideas in a public setting, such as on social media.

3. A statement, or set of statements, that describes general principles about how variables relate to one another is a(n)

1. prediction

2. hypothesis

3. empirical observation

4. theory

4. Why is publication an important part of the research process?

1. Because publication enables practitioners to use the research in applied settings.

2. Because when a study is published, other scientists can verify or challenge it, making science self-correcting.

3. Because journalists can make the knowledge available to the general public.

4. Because publication is the first step of the theory-data cycle.

5. Which of the following research questions best illustrates an example of basic research?

1. Has our company's new marketing campaign led to an increase in sales?

2. How satisfied are our patients with the sensitivity of the nursing staff?

3. Does wearing kinesio-tape reduce joint pain in figure skaters?

4. Can 2-month-old human infants tell the difference between four objects and six objects?

Lecture 10:

1. Which of the following variables are manipulated, rather than measured? (Could be more than one.)

1. Number of shoes owned, in pairs.

2. A person's height, in cm.

3. Amount of aspirin a researcher gives a person to take, either 325 mg or 500 mg.

4. Degree of happiness, rated on a scale from 1 to 10.

5. Type of praise a researcher uses in groups of dogs: verbal praise or a clicking sound paired with treats.

2. Which of the following headlines is an association claim?

1. Chewing Gum Can Improve Your Mood and Focus

2. Handling Money Decreases Helpful Behavior in Young Children

3. Workaholism Is Tied to Psychiatric Disorders

4. Eating Kiwis May Help You Fall Asleep

3. Which of the following headlines is a frequency claim?

1. Obese Kids Are Less Sensitive to Tastes

2. Eighty Percent of Women Feel Dissatisfied with How Their Bodies Look

3. Feeling Fat? Maybe Facebook Is to Blame

4. Daycare and Behavior Problems Are Not Linked

4. Which of the following headlines is a causal claim?

1. Taking a Deep Breath Helps Minimize High Blood Pressure, Anxiety, and Depression

2. Younger People Can't Read Emotions on Wrinkled Faces

3. Strange But True: Babies Born in the Autumn Are More Likely to Live to 100

4. Check the Baby! Many New Moms Show Signs of OCD

5. Which validity would you be interrogating by asking: How well did the researchers measure sensitivity to tastes in this study?

1. Construct validity

2. Statistical validity

3. External validity

4. Internal validity

6. Which validity would you be interrogating by asking: How did the researchers get their sample of people for this survey?

1. Construct validity

2. Statistical validity

3. External validity

4. Internal validity

7. In most experiments, trade-offs are made between validities because it is not possible to achieve all four at once. What is the most common trade- off?

1. Internal and external validity.

2. Construct and statistical validity.

3. Statistical and internal validity.

4. External and statistical validity

8. A regression threat applies especially:

1. When there are two groups in the study: an experimental group and a control group.

2. When the researcher recruits a sample whose average is extremely low or high at pretest.

3. In a posttest-only design.

4. When there is a small sample in the study.

9. Dr. Banks tests to see how many training sessions it takes for dogs to learn to "Sit and stay." She randomly assigns 60 dogs to two reward conditions: one is miniature hot dogs, the other is small pieces of steak. Surprisingly, she finds the dogs in each group learn "Sit and stay" in about the same number of sessions. Given the design of her study, what is the most likely explanation for this null effect?

1. The dogs loved both treats (her reward manipulation has a ceiling effect).

2. She used too many dogs.

3. She didn't use a manipulation check.

4. There were too many individual differences among the dogs.

Lecture 11:

1. The following item appears on a survey: "Was your cell phone purchased within the last two years and have you downloaded the most recent updates?" What is the biggest problem with this wording? 1. It is a leading question.

2. It involves negative wording.

3. It is a double-barreled question.

4. It is not on a Likert scale.

2. When people are using an acquiescent response set they are:

1. Trying to give the responses they think the researcher wants to hear.

2. Misrepresenting their views to appear more socially acceptable.

3. Giving the same, neutral answer to each question.

4. Tending to agree with every item, no matter what it says.

Lecture 12:

1. Suppose you hear that conscientious people are more likely to get regular health checkups. Which of the following correlations between conscientiousness and getting checkups would probably support this claim?

1. r = .03

2. r = .45

3. r = -.35

4. r = -1.0

Lecture 13:

Lecture 14:

1. What is the term for a quasi-experimental design with at least one treatment group and one comparison group, in which the participants are measured once and have not been randomly assigned to the groups?

1. Nonequivalent control group design

2. Independent-groups design

3. Factorial design

4. Reversal design

2. Which of these is not a reason for a researcher to select a quasiexperimental design?

1. To enhance external validity.

2. To avoid the ethical issues a true experiment would cause.

3. To ensure internal validity.

4. To take advantage of real-world opportunities to study phenomena and events.

3. Researchers studied preteens at an outdoor education camp that prevented kids from using devices with screens, such as cell phones (Uhls et al., 2014). They predicted the camp would improve children's nonverbal communication skills. One group of sixth graders attended the camp in the spring. They were compared with a group of sixth graders from the same school who had not attended the camp. The kids were tested on how well they could read emotions in faces. The camp group took the emotion test on Monday and Friday (the first and last days of their camp week); the control group took the emotion test on Monday and Friday of a regular school week. What type of design was this?

1. Interrupted time-series design

2. Nonequivalent control group interrupted time-series design

3. Nonequivalent control group posttest-only design

4. Nonequivalent control group pretest/posttest design

4. For the kids' camp study, you could ask: Did the researchers use a reliable and valid test of nonverbal communication skills? You would be asking about which kind of validity?

1. Construct validity

2. Statistical validity

3. Internal validity

4. External validity